Two symmetries of the circle and an elliptic angle

a line integral evaluated without ever computing it

Problem. Let $D = \{x^2 + y^2 < 1\}$ and let $\ell = \partial D$ be traversed once counterclockwise. Prove

$$\oint_\ell \frac{x\mathrm{e}^{\sin y}\,\mathrm{d}y - y\mathrm{e}^{-\sin x}\,\mathrm{d}x}{4x^2+5y^2} = \oint_\ell \frac{x\mathrm{e}^{-\sin y}\,\mathrm{d}y - y\mathrm{e}^{\sin x}\,\mathrm{d}x}{5x^2+4y^2}, \tag{I}$$

$$\oint_\ell \frac{x\mathrm{e}^{\sin y}\,\mathrm{d}y - y\mathrm{e}^{-\sin x}\,\mathrm{d}x}{4x^2+5y^2} > \frac{53\pi}{120}. \tag{II}$$

(I) by the reflection $(x,y)\mapsto(y,x)$; (II) by the antipode $(x,y)\mapsto(-x,-y)$ and $\oint_\ell \frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{4x^2+5y^2} = \frac{\pi}{\sqrt{5}}$.

Overview. Neither integral can be evaluated in closed form, and neither needs to be. Two isometries of the circle do everything.

The reflection $S(x,y) = (y,x)$ maps ℓ to itself with reversed orientation and carries the second differential form to minus the first; the two sign changes cancel, which is (I).

The antipodal map $A(x,y) = (-x,-y)$ preserves ℓ *with* its orientation and replaces e^u by e^{-u}. Averaging the form with its pullback therefore turns both exponentials into cosh, and $\cosh \geq 1$ gives a clean lower bound by the “elliptic angle form” $\frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{4x^2+5y^2}$. That form is $\frac{1}{2\sqrt{5}}\,\mathrm{d}\,\mathrm{Arg}(2x + \mathrm{i}\sqrt{5}y)$, and the image of the unit circle under $(x,y)\mapsto(2x,\sqrt{5}y)$ winds once around the origin, so the integral is $\frac{2\pi}{2\sqrt{5}} = \frac{\pi}{\sqrt{5}}$. Finally $\frac{1}{\sqrt{5}} > \frac{53}{120}$.

Write

$$\omega = \frac{x\mathrm{e}^{\sin y}\,\mathrm{d}y - y\mathrm{e}^{-\sin x}\,\mathrm{d}x}{4x^2+5y^2}, \qquad \eta = \frac{x\mathrm{e}^{-\sin y}\,\mathrm{d}y - y\mathrm{e}^{\sin x}\,\mathrm{d}x}{5x^2+4y^2},$$

and $I = \oint_\ell \omega$, $J = \oint_\ell \eta$.

1. The integrals in parametric form

With the standard positive parametrisation

$$x = \cos t, \quad y = \sin t, \quad \mathrm{d}x = -\sin t\,\mathrm{d}t, \quad \mathrm{d}y = \cos t\,\mathrm{d}t, \qquad 0 \leq t \leq 2\pi,$$

abbreviate $c = \cos t$, $s = \sin t$. Then

$$x\mathrm{e}^{\sin y}\,\mathrm{d}y = c^2\mathrm{e}^{\sin s}\,\mathrm{d}t, \qquad -y\mathrm{e}^{-\sin x}\,\mathrm{d}x = s^2\mathrm{e}^{-\sin c}\,\mathrm{d}t,$$

so

$$I = \int_0^{2\pi} \frac{c^2\mathrm{e}^{\sin s} + s^2\mathrm{e}^{-\sin c}}{4c^2+5s^2}\,\mathrm{d}t, \qquad J = \int_0^{2\pi} \frac{c^2\mathrm{e}^{-\sin s} + s^2\mathrm{e}^{\sin c}}{5c^2+4s^2}\,\mathrm{d}t. \tag{1}$$

Both integrands are continuous and positive, so $I, J > 0$; the denominators vanish only at the origin, which is not on ℓ.

2. Part (I): the reflection $x \leftrightarrow y$

Let $S(x,y) = (y,x)$, the reflection in the diagonal. It maps the unit circle onto itself but reverses orientation:

$$S(\ell) = -\ell. \tag{2}$$

Its pullback substitutes $x \mapsto y,\ y \mapsto x,\ \mathrm{d}x \mapsto \mathrm{d}y,\ \mathrm{d}y \mapsto \mathrm{d}x$ in η:

$$S^*\eta = \frac{y\mathrm{e}^{-\sin x}\,\mathrm{d}x - x\mathrm{e}^{\sin y}\,\mathrm{d}y}{5y^2+4x^2} = -\frac{x\mathrm{e}^{\sin y}\,\mathrm{d}y - y\mathrm{e}^{-\sin x}\,\mathrm{d}x}{4x^2+5y^2} = -\omega. \tag{3}$$

The change-of-variables rule for line integrals gives $\int_{S(\ell)}\eta = \int_\ell S^*\eta$, so by (2) and (3)

$$-\oint_\ell \eta = -\oint_\ell \omega \qquad \Longrightarrow \qquad \boxed{I = J.}$$

The same statement without the language of forms: in (1) substitute $t \mapsto \frac{\pi}{2} - t$, which exchanges cos and sin. The denominator $5c^2+4s^2$ becomes $4c^2+5s^2$, the numerator of J becomes the numerator of I, and the two integrals over the full period coincide.

3. Why Green's theorem is the wrong first move

The natural reflex is to apply Green's theorem to ω over D. It does not apply: the denominator $4x^2+5y^2$ vanishes at the origin, which lies inside D, so the field is not C^1 on the region.

One can excise a small disk around the origin, apply Green's theorem to the annulus, and let the radius tend to zero. But then the boundary term at the small circle must be evaluated, and it is exactly the quantity one is after. The symmetries below are cleaner because they never leave the circle.

4. Part (II): the antipodal map produces cosh

Let $A(x,y) = (-x,-y)$, the rotation by π. It preserves the unit circle *and* its orientation:

$$A(\ell) = \ell.$$

Since sin is odd and the denominator is even,

$$A^*\omega = \frac{x\mathrm{e}^{-\sin y}\,\mathrm{d}y - y\mathrm{e}^{\sin x}\,\mathrm{d}x}{4x^2+5y^2}, \tag{4}$$

and $\oint_\ell \omega = \oint_\ell A^*\omega$. Averaging,

$$I = \frac{1}{2}\oint_\ell \big(\omega + A^*\omega\big), \qquad \frac{\omega + A^*\omega}{2} = \frac{x\cosh(\sin y)\,\mathrm{d}y - y\cosh(\sin x)\,\mathrm{d}x}{4x^2+5y^2}, \tag{5}$$

using $\frac{\mathrm{e}^u+\mathrm{e}^{-u}}{2} = \cosh u$.

On the unit circle both differentials appearing in (5) are *non-negative*:

$$x\,\mathrm{d}y = c^2\,\mathrm{d}t \geq 0, \qquad -y\,\mathrm{d}x = s^2\,\mathrm{d}t \geq 0.$$

Therefore $\cosh \geq 1$ may be applied termwise:

$$I = \int_0^{2\pi} \frac{c^2\cosh(\sin s) + s^2\cosh(\sin c)}{4c^2+5s^2}\,\mathrm{d}t \;\geq\; \int_0^{2\pi} \frac{c^2+s^2}{4c^2+5s^2}\,\mathrm{d}t = \oint_\ell \frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{4x^2+5y^2}. \tag{6}$$

This is not a pointwise bound on the original integrand. At individual points the integrand of I is smaller than the base integrand, because $\mathrm{e}^{-\sin c} < 1$ wherever $\sin c > 0$. The inequality appears only after averaging the values at the antipodal pair t and $t+\pi$, where the two exponentials are reciprocals and their mean is a cosh.

5. The base integral is an angle

Let

$$K = \oint_\ell \frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{4x^2 + 5y^2}.$$

Apply the linear stretch

$$u = 2x, \qquad v = \sqrt{5}\,y, \qquad W = u + \mathrm{i}v = 2x + \mathrm{i}\sqrt{5}\,y,$$

so that $u^2 + v^2 = 4x^2 + 5y^2$. For any $W \neq 0$,

$$\mathrm{d}\,\mathrm{Arg}\,W = \mathrm{Im}\,\frac{\mathrm{d}W}{W} = \frac{u\,\mathrm{d}v - v\,\mathrm{d}u}{u^2 + v^2},$$

and here $\mathrm{d}u = 2\,\mathrm{d}x,\ \mathrm{d}v = \sqrt{5}\,\mathrm{d}y$, so

$$u\,\mathrm{d}v - v\,\mathrm{d}u = 2\sqrt{5}\big(x\,\mathrm{d}y - y\,\mathrm{d}x\big).$$

Hence

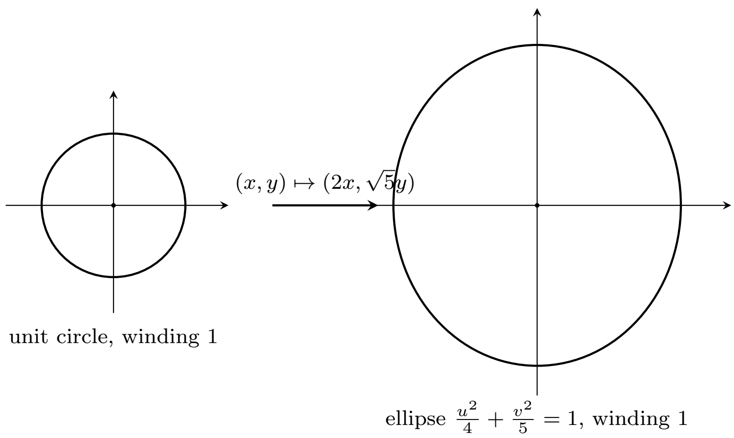
$$\boxed{\frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{4x^2 + 5y^2} = \frac{1}{2\sqrt{5}}\,\mathrm{d}\,\mathrm{Arg}\big(2x + \mathrm{i}\sqrt{5}\,y\big).} \tag{7}$$

As (x, y) traverses the unit circle once, the point

$$W(t) = 2\cos t + \mathrm{i}\sqrt{5}\sin t$$

traverses an ellipse once around the origin, so its argument increases by 2π and

$$K = \frac{1}{2\sqrt{5}} \cdot 2\pi = \frac{\pi}{\sqrt{5}}. \tag{8}$$



Geometrically, $\frac{x\,\mathrm{d}y - y\,\mathrm{d}x}{x^2+y^2}$ measures the increment of the ordinary polar angle. Replacing the denominator by $4x^2 + 5y^2$ measures instead the angle *after* the stretch. The stretch turns the circle into an ellipse, but the ellipse still encircles the origin exactly once, since winding number is a topological invariant, so the total increment is again 2π, only rescaled by $\frac{1}{2\sqrt{5}}$.

A direct check confirms (8): with $c^2 + s^2 = 1$,

$$K = \int_0^{2\pi} \frac{\mathrm{d}t}{4\cos^2 t + 5\sin^2 t} = \frac{2\pi}{2 \cdot \sqrt{5}} = \frac{\pi}{\sqrt{5}},$$

by the standard formula $\int_0^{2\pi} \frac{\mathrm{d}t}{a^2\cos^2 t + b^2 \sin^2 t} = \frac{2\pi}{ab}$ with $a = 2$, $b = \sqrt{5}$, itself just (7) in disguise.

6. Conclusion of part (II)

From (6) and (8),

$$I \;\geq\; \frac{\pi}{\sqrt{5}}.$$

It remains to compare with $\frac{53\pi}{120}$. Since all quantities are positive,

$$\frac{1}{\sqrt{5}} > \frac{53}{120} \iff 120^2 > 5\cdot 53^2 \iff 14400 > 14045,$$

which is true. Hence

$$\boxed{I \;\geq\; \frac{\pi}{\sqrt{5}} = 1.4049629\ldots \;>\; \frac{53\pi}{120} = 1.3875368\ldots,}$$

which is (II).

The true value, by numerical quadrature of (1), is

$$I = J = 1.5580643\ldots,$$

so the chain of bounds is comfortable: the $\cosh \geq 1$ step costs about 10%, and the rational comparison a further 1.3%.

Quantity	Exact	Value
The integral $I = J$	–	1.5580643
After $\cosh \geq 1$	$\pi/\sqrt{5}$	1.4049629
Target	$53\pi/120$	1.3875368

7. The elliptic angle as a Blaschke product

This section is not needed for the proof; it identifies the complex object behind the stretch.

On $|z| = 1$, with $z = x + \mathrm{i}y$, one has $\bar{z} = z^{-1}$ and

$$x = \frac{z + z^{-1}}{2}, \qquad y = \frac{z - z^{-1}}{2\mathrm{i}},$$

so

$$W = 2x + \mathrm{i}\sqrt{5}\,y = \frac{2+\sqrt{5}}{2}\,z + \frac{2-\sqrt{5}}{2}\,z^{-1} = \frac{2+\sqrt{5}}{2}\,z^{-1}\big(z^2 - \rho\big), \tag{9}$$

where

$$\rho = -\frac{2-\sqrt{5}}{2+\sqrt{5}} = 9 - 4\sqrt{5} = (\sqrt{5}-2)^2 = 0.0557\ldots < 1.$$

Formula (9) makes the winding number transparent: on $|z| = 1$ the factor z^{-1} contributes winding -1, while $z^2 - \rho$ has its two zeros $z = \pm\sqrt{\rho}$ inside the disk and contributes $+2$. The total is $+1$, confirming $\Delta\,\mathrm{Arg}\,W = 2\pi$ and hence (8).

Moreover $\rho \in (0,1)$ means

$$B(w) = \frac{w - \rho}{1 - \rho w}$$

is a Blaschke factor of the disk, and $|z^2 - \rho| = |1 - \rho z^2|$ on $|z| = 1$; the elliptic angle differs from the circular one by the argument of a Blaschke product in z^2. The eccentricity of the ellipse and the modulus ρ of that Blaschke factor are two names for the same number: $\rho = \dfrac{b-a}{b+a}$ with $a = 2$, $b = \sqrt{5}$, since $\dfrac{\sqrt{5}-2}{\sqrt{5}+2} = (\sqrt{5}-2)^2 = 9 - 4\sqrt{5}$.

8. What generalises

The proof used almost nothing about the specific functions.

Ingredient	Property used
$e^{\sin y}$, $e^{-\sin x}$	That the two exponents are odd and opposite, so that the antipodal average is $\cosh \geq 1$. Any $e^{g(y)}, e^{-g(x)}$ with g odd works.
The denominators $4x^2 + 5y^2$ and $5x^2 + 4y^2$	That they are exchanged by $x \leftrightarrow y$ (part I), and that the first is a positive quadratic form (part II).
The coefficients $4, 5$	Only through $\sqrt{4 \cdot 5} = 2\sqrt{5}$ in the final constant $\frac{\pi}{\sqrt{5}} = \frac{2\pi}{2\sqrt{5}}$.
The unit circle	That it is invariant under both S and A, and that it winds once around the origin.

Thus for any odd g, any $a, b > 0$, and ℓ the unit circle,

$$\oint_\ell \frac{x e^{g(y)}\, dy - y e^{-g(x)}\, dx}{a^2 x^2 + b^2 y^2} \;\geq\; \frac{2\pi}{ab},$$

with equality only for $g \equiv 0$. The stated problem is the case $g = \sin$, $a = 2$, $b = \sqrt{5}$.